Ahmed Ayman Al-Hofy

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https://github.com/AhmedAyman4 | https://www.linkedin.com/in/ahmed-alhofy/ | View my Portfolio!

Summary

AI Engineer and **Data Scientist** with a background in **Information Systems**, focused on **RAG**, **NLP**, and **data automation**. Experienced in building **end-to-end agents** and **predictive models** that solve **real-world problems**. Passionate about bridging the gap between **raw data** and **functional AI products**.

Education

Bachelor's Degree, Faculty of Computers & Artificial Intelligence – USC — Graduating June 2026

CGPA: 3.36/4.0 | Major: Information Systems.

Experience & Training

AI Engineer | Konecta · Internship — New Cairo, Egypt | Jul 2025 – Jan 2026

- Collected, cleaned, and structured data using Python and web scraping tools to support reliable analysis and modeling.
- Built and optimized machine learning models (classification, clustering, image recognition) and implemented RAG-based chatbots with embeddings, document indexing, and multi-turn conversations.
- Automated data and AI workflows using n8n, improving efficiency and reducing manual processing.

Data scientist | Digital Egypt Pioneers Initiative - DEPI · Internship — Alexandria, Egypt | Oct 2024 – May 2025

- Learned to analyze data using Python, SQL, and analytics tools to uncover trends, opportunities, and actionable business insights.
- Learned to build, deploy, and track ML models using Python, Scikit-learn, and MLflow, enhanced with AI and prompt engineering for seamless integration.

Intern | CIB Egypt · Internship — Cairo, Egypt | Jul 2024 - Jul 2024

- Developed data, financial, and entrepreneurial skills to strengthen problem-solving and decision-making.

Skills

Python (ML/DS) | SQL | Bash | HTML/CSS | LangChain | LangGraph | LLMs (Hugging Face) | Prompt Engineering | RAG | Matplotlib | Seaborn | Scikit-learn | TensorFlow | Web Scraping (Playwright, BeautifulSoup) | AWS | Google Cloud Platform (GCP) | Vector Databases | Stats | Tableau | Power BI | Excel | Docker | Git | OOP | Problem-Solving | Communication

Projects

Nawy Property Recommender (AI Real Estate Consultant) — GitHub | Live Demo

- Developed an AI property consultant using a RAG pipeline (LangChain, LangSmith) to enable efficient querying of nearly 9,000 properties across 1,700+ compounds and 40+ locations, optimizing system latency and operational costs.
- Engineered an intelligent routing system utilizing Vector DB semantic search and LLM-based filtering to deliver precise, high-intent property recommendations and automated market comparisons.
- Built a scalable full-stack platform (FastAPI, Docker, Next.js) featuring an XGBoost model trained on 29,000+ scraped listings to provide real-time property price predictions and market insights.

Semantic Book Recommender — GitHub | Live Demo

- Accomplished a 40% improvement in recommendation relevance (compared to keyword-based search) by engineering a semantic search engine using Sentence-Transformers for vector embeddings and Zero-Shot classification to map books to specific emotional and categorical tags.
- Developed and deployed a production-ready full-stack application with a Next.js frontend and FastAPI backend, achieving seamless multi-environment scalability through Docker containerization and automated GitHub Actions workflows.
- Optimized data discovery and user insight by preprocessing a dataset of 7,000+ books through comprehensive EDA, NLP-driven sentiment analysis, and the creation of custom-tagged datasets in Jupyter Notebooks, enabling users to find titles based on mood and complex semantic queries.

Customer Segmentation — GitHub | Live Demo

- Segmented wholesale customers using K-Means, PCA, and rigorous evaluation to define six actionable profiles, driving targeted marketing and inventory optimization.

Automated Document Processing: — GitHub | Live Demo

- Built an AI document processor using OCR, LangChain, and LLM to extract structured data, cutting manual work by 80%; deployed a Streamlit app with MongoDB search for real-time querying.

Courses And Certifications

- SAP Certified Associate - SAP Generative AI Developer Issued by SAP — Sep 2025
- AWS Certified AI Practitioner Issued by Amazon Web Services Training and Certification — Apr 2025
- Oracle Cloud Infrastructure 2025 Certified Al Foundations Associate Issued by Oracle — Jun 2025
- Data Scientist Associate Issued by Data Camp — Aug 2024
- IBM SkillsBuild Data Analytics Certificate Issued by IBM SkillsBuild — Jun 2025
- AWS Academy Graduate - AWS Academy Machine Learning for Natural Language Processing — Aug 2024
- AWS Academy Graduate - AWS Academy Cloud Foundations — Jun 2024
- AWS Academy Graduate - AWS Academy Data Engineering — Jun 2024
- Applied Data Science with Python - Level 2 Issued By IBM — May 2024
- Google Data Analytics Professional Certificate Issued by Coursera — Jul 2024
- The Machine Learning Process A-Z – Issued By 365 Data Science — Nov 2024
- McKinsey Forward Program - Issued by McKinsey & Company — Dec 2024